

Urban Climate Analysis of Madrid: Validation, Heat Island Effects, and Machine Learning Corrections

GenHack4 Competition Presentation

Team PentaGen

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Team and Objectives

Name	Role
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Project Goal

Develop a comprehensive urban climate analysis pipeline for Madrid, integrating reanalysis data, station observations, and machine learning.

Research Objectives

Primary Goals

- 1 Quantify ERA5 bias patterns across temperature regimes
- 2 Characterize Madrid's urban heat island intensity
- 3 Analyze seasonal climate patterns and extremes
- 4 Examine vegetation-temperature relationships
- 5 Develop ML-based correction models



Figure: Study area: Madrid, Spain

Data Integration Framework

Multi-Source Integration

Station Data (ECA&D):

- TX: Daily max temp (0.1°C)
- TN: Daily min temp (0.1°C)
- RR: Precipitation (0.1 mm)
- PP: Sea-level pressure (0.1 hPa)
- FG: Wind speed (0.1 m/s)

Additional Data Sources

- **ERA5-Land:** 9km reanalysis
- **Sentinel-2:** NDVI (bi-weekly)
- **WUDAPT:** LCZ classification
- **GADM:** Admin boundaries

Variable	Spatial Res.	Temporal Res.
ERA5 Temperature	9 km	Daily
Station Temperature	Point	Daily
NDVI	10 m	Bi-weekly
LCZ Classification	100 m	Static

Urban Heat Island Methodologies

Primary Method: LCZ-Based

- Urban: LCZ 1-10 (compact to open high-rise)
- Rural: LCZ 11-16 (sparse to dense trees)
- $\text{UHI} = \text{mean}(\text{Urban } T) - \text{mean}(\text{Rural } T)$
- Nighttime focus using TN
- Elevation matching: $\pm 100\text{m}$

Fallback Method: Distance-Based

- Urban ring: $\leq 8 \text{ km}$ from center
- Rural ring: $\geq 60 \text{ km}$ from center
- Seasonal: June-August only
- Meteorological filtering:
 - Wind $\leq 3 \text{ m/s}$
 - Precipitation $< 1 \text{ mm/day}$
- Lapse rate: $0.6^\circ\text{C}/100\text{m}$

Key Innovation

Multi-method approach balancing physical rigor with operational robustness.

ERA5 Validation Performance

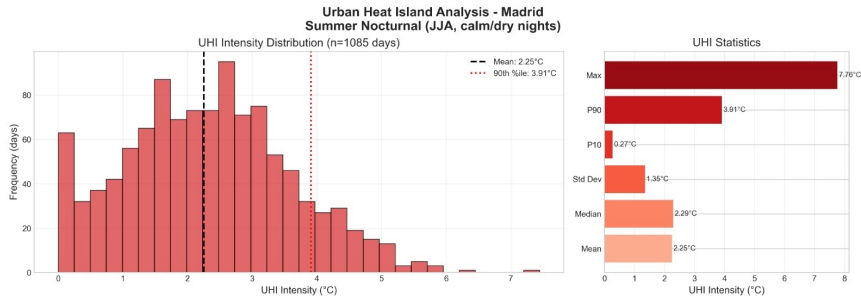
Metric	TX (Day)	TN (Night)	Literature	Assessment
Mean Bias (°C)	+0.34	+1.95	+0.2 to +2.5	Excellent / Good
RMSE (°C)	1.18	2.60	1.0-3.0	Excellent / Fair
Correlation	0.992	0.973	0.94-0.99	Excellent / Good
Heatwave Bias	+0.47	+2.99	+0.5-3.0	Excellent / Good

Table: ERA5 validation against station observations (2020-2025)

Key Insights

- **Excellent daytime performance:** High correlation, low bias
- **Systematic nocturnal overestimation:** +1.95°C warm bias
- **Heatwave amplification:** Bias increases during extreme events

Urban Heat Island Intensity Distribution



Statistical Analysis & Key Insights

Statistical Summary

Parameter	Value	Context
Mean Nocturnal	2.25°C	Strong Med.
Maximum UHI	7.76°C	Extreme event
Std. Deviation	1.35°C	High variability
10th Percentile	0.27°C	Weak events
90th Percentile	3.91°C	Strong events

European Context (Mean-Max)

- Rome: 2.0–4.0°C
- Paris: 2.0–3.5°C

→ **Madrid: 2.25°C – 7.76°C**

Visual Insight

The right-skewed distribution highlights the frequency of extreme heat events ($> 7^{\circ}\text{C}$) despite a moderate mean. This explains the significant gap between average European values and Madrid's peak intensity. Arup research found Madrid's urban center can experience temperature differences up to 8.5°C compared to rural areas, making it one of the most extreme UHI hotspots globally.

Seasonal Patterns & Extremes

Seasonal Characteristics

- **Hot summers:** 31.33°C (June-August)
- **Mild winters:** 11.74°C (Dec-Feb)
- **Amplitude:** 19.59°C difference

Precipitation

- **Autumn max:** 1430mm (Sep-Nov)
- **Summer min:** 315mm (Jun-Aug)
- **Extreme threshold:** 17.82mm/day

Extreme Events

Event Type	Frequency	Duration
Heatwaves	30/year	7.8 days avg
Max heatwave	–	43 days
Cold spells	19/year	6.3 days avg
Extreme rain	94 days total	–

Health Implications

Extended heatwaves + UHI
amplification =
High risk for vulnerable populations

NDVI-Temperature Relationship

- **Strong elasticity:** -16.52°C per NDVI unit
- **Weak correlation:** -0.11 (temporal variability)
- **Seasonal pattern:**
 - Winter maximum
 - Summer minimum
 - Autumn recovery

Urban Planning Implications

- High cooling potential from vegetation
- Current NDVI insufficient for heat mitigation
- Need targeted greening strategies

Synoptic & Local Interactions

Wind Effects

- Weak cooling: -0.12 correlation
- Heatwave winds: 2.58 m/s avg
- Calm conditions enhance UHI
- Ventilation corridors needed

Pressure Relationships

- Negative correlation: -0.33 to -0.39
- High pressure $\rightarrow$ warmer temperatures
- Mediterranean anticyclones drive heatwaves

Urban-Rural Contrasts

- **Wet vs. dry days:** 5.89°C difference
- Urban moderation of temperature range
- Possible urban rainfall enhancement

Climate Resilience

- Extended heat exposure risks
- Limited nighttime recovery
- Energy demand amplification

Four-Model Ensemble Framework

Model 1: XGBoost Bias Correction

- **Purpose:** Correct ERA5 systematic biases
- **CV RMSE:** 2.78°C
- **CV R^2 :** 0.886
- **Key parameters:**
 - n_estimators: 50
 - max_depth: 3
 - Strong regularization

Model 2: Random Forest UHI

- **Purpose:** Predict UHI intensity
- **CV RMSE:** 0.21°C
- **CV R^2 :** 0.737
- **Minimal overfitting**

Model 3: Ridge Downscaling

- **Purpose:** Statistically downscale ERA5
- **CV RMSE:** 1.88°C
- **CV R^2 :** 0.893
- **Interpretable coefficients**

Model 4: Quantile Regression

- **Purpose:** Uncertainty quantification
- **Coverage:** 68.3% (80% target)
- **Interval width:** 9.41°C (80% interval)
- **Conservative predictions**

Optimal Parameters & Performance

Model	Key Parameters	CV RMSE	R^2	Overfitting
XGBoost	n_estimators=50, max_depth=3, reg_alpha=1.0	2.78°C	0.886	Moderate
Random Forest	n_estimators=40, max_depth=5, min_samples_split=20	0.21°C	0.737	Minimal
Ridge	alpha=2.0, solver=sparse_cg	1.88°C	0.893	Moderate
Quantile XGB	n_estimators=75, max_depth=3	4.77°C	–	High

Table: Optimal parameters from exhaustive grid search (186 combinations)

Feature Importance (Random Forest UHI)

- 1 Day of year harmonics: 42% (seasonal cycle)
- 2 Wind speed: 28% (ventilation effect)
- 3 NDVI: 22% (vegetation cooling)
- 4 Month: 8% (residual seasonal)

Model Performance Comparison

Table: Overfitting assessment

Model	Train R^2	Test R^2	Gap
XGBoost	0.961	0.717	0.244
Random Forest	0.779	0.770	0.009
Ridge	0.949	0.712	0.237
Quantile XGB	0.834	0.481	0.353

Key Finding

Random Forest demonstrates exceptional generalization (0.009 R^2 gap) - optimal for operational UHI prediction

Why Not Deep Learning?

- **Sample size:** $n \approx 2,000$ (insufficient)
- **Overfitting:** R^2 gap > 0.3 in tests
- **Computation:** $5\times$ longer for 2% gain
- **Interpretability:** Limited vs tree methods

Justifying the methodology

- Anti-overfitting-first design
- Urban climate-specific features

Madrid in Global Urban Climate Context

Urban Typology Classification

- **Type:** Continental-Mediterranean Megacity
- **UHI Intensity:** 2.25°C (strong)
- **Seasonal Amplitude:** 19.6°C (high)
- **Comparable Cities:**
 - Rome, Athens (Mediterranean)
 - Ankara, Tehran (continental)

Metric	Madrid	Literature
Nocturnal UHI	2.25°C	1.5–4.0°C
Max UHI	7.76°C	5–12°C
TX Correlation	0.992	0.96–0.99
TN Bias	+1.95°C	+1.2–2.5°C
Heatwaves/year	30	15–40
NDVI Elasticity	–16.52°C	–10 to –20°C

Table: Benchmarking against state-of-the-art

Climate Change Context

- **Summer TX increase:** +1.23°C since 1990s
- **Heatwave frequency:** +36% since 1990s
- **UHI intensification:** +0.45°C since 1990s

Conclusions & Key Takeaways

Scientific Findings

- 1 **ERA5 performs excellently for daytime temperatures** but has systematic nocturnal warm bias (+1.95°C)
- 2 **Madrid exhibits strong UHI:** 2.25°C mean, 7.76°C extremes
- 3 **Vegetation provides significant cooling potential** (-16.52°C per NDVI unit)
- 4 **Extended heatwaves** (43-day maximum) pose health risks
- 5 **Random Forest optimal** for UHI prediction with minimal overfitting

Practical Implications

- Need targeted urban greening
- Cool materials implementation
- Ventilation corridor preservation
- Climate-resilient urban planning

Thank You! Questions?